

Grade 7 Mathematics

Review Pack

Lebanese Curriculum | English

Clear explanations | Worked examples | Practice | Reviews | Answer key

Independent educational material prepared with reference to the published CRDP curriculum.

Not an official Ministry of Education or CRDP publication.

How to Use This Pack

This pack reviews the Grade 7 mathematics topics listed in the English scope and sequence published by Lebanon's Centre for Educational Research and Development (CRDP). Work through one lesson at a time, cover the worked example, and try the questions before checking the answer key.

Study routine

- Read the short explanation and copy the rule in your own words.
- Follow the worked example one step at a time.
- Solve the practice questions on paper without looking at the answers.
- Correct your work in a different colour and redo any missed question the next day.

Topics in this pack

1. Prime Numbers and Prime Factorization
2. Reducing Fractions
3. Writing Fractions as Decimals
4. Subtracting and Multiplying Integers
5. Powers with Positive Integer Exponents
6. Common Factors and Factorization
7. Direct Proportion
8. Calculating with Algebraic Expressions
9. Equations of the Form $ax = b$
10. Coordinates and Geometric Loci
11. Constructing Perpendicular and Angle Bisectors
12. Congruent Triangles and Angles on Parallel Lines
13. Representing Cubes and Rectangular Prisms
14. Translations
15. Relative Frequencies and Graphs

Scope note

The pack is an independent review resource. Schools may sequence topics differently or use a yearly pacing plan. Follow your teacher's assigned chapters when preparing for class assessments.

Lesson 1 Prime Numbers and Prime Factorization

Recognize prime and composite numbers and write a whole number as a product of prime factors.

Learn the idea

A prime number is a whole number greater than 1 with exactly two positive factors: 1 and itself. The number 1 is neither prime nor composite.

To factorize a number, divide it by prime numbers until every factor is prime. Use exponents to show repeated factors.

Worked example

$$180 = 2 \times 90 = 2 \times 2 \times 45 = 2^2 \times 3^2 \times 5.$$

Try it

1. Classify each number as prime, composite, or neither: 1, 2, 15, 23.

Answer: _____

2. Write 72 as a product of prime factors.

Answer: _____

3. Write 105 as a product of prime factors.

Answer: _____

4. Is 91 prime? Show a factor pair.

Answer: _____

Lesson 2 Reducing Fractions

Write a fraction in its simplest form by dividing its numerator and denominator by the same common factor.

Learn the idea

A fraction keeps the same value when its numerator and denominator are divided by the same non-zero number.

Use the greatest common factor to reduce a fraction in one step. A fraction is in simplest form when the numerator and denominator have no common factor greater than 1.

Worked example

Reduce $54/72$. The greatest common factor is 18, so $54/72 = (54 \div 18)/(72 \div 18) = 3/4$.

Try it

1. Reduce $21/35$.

Answer: _____

2. Reduce $48/60$.

Answer: _____

3. Reduce $63/81$.

Answer: _____

4. Reduce $45/120$.

Answer: _____

Lesson 3 Writing Fractions as Decimals

Convert a fraction to a decimal by making an equivalent fraction with denominator 10, 100, or 1,000, or by dividing.

Learn the idea

If the denominator can be changed to 10, 100, or 1,000, use an equivalent fraction. Otherwise, divide the numerator by the denominator.

Keep the place value in mind: tenths, hundredths, and thousandths.

Worked example

$7/20 = 35/100 = 0.35$. Also, $3/8 = 375/1,000 = 0.375$.

Try it

1. Write $3/8$ as a decimal.

Answer: _____

2. Write $9/25$ as a decimal.

Answer: _____

3. Write $7/4$ as a decimal.

Answer: _____

4. Write $11/20$ as a decimal.

Answer: _____

Lesson 4 Subtracting and Multiplying Integers

Subtract integers and multiply integers using sign rules.

Learn the idea

Subtracting a number is the same as adding its opposite: $a - b = a + (-b)$. In particular, subtracting a negative becomes addition.

When multiplying two integers with the same sign, the product is positive. When their signs differ, the product is negative.

Worked example

$$14 - (-6) = 14 + 6 = 20. \quad (-7) \times 6 = -42.$$

Try it

1. Calculate $14 - (-6)$.

Answer: _____

2. Calculate $-9 - 4$.

Answer: _____

3. Calculate $(-7) \times 6$.

Answer: _____

4. Calculate $(-12) \times (-3)$.

Answer: _____

Lesson 5 Powers with Positive Integer Exponents

Read and evaluate a power whose base is positive and whose exponent is a positive integer.

Learn the idea

For a positive number a and positive integer n , a^n means multiply a by itself n times. For example, $3^4 = 3 \times 3 \times 3 \times 3$.

An exponent tells how many equal factors there are; it does not mean multiply the base by the exponent.

Worked example

$2^5 = 2 \times 2 \times 2 \times 2 \times 2 = 32$. $(\frac{3}{2})^2 = \frac{9}{4}$.

Try it

1. Evaluate 4^3 .

Answer: _____

2. Evaluate 7^2 .

Answer: _____

3. Evaluate $(\frac{1}{2})^3$.

Answer: _____

4. Calculate $3^2 + 2^3$.

Answer: _____

Lesson 6 Common Factors and Factorization

Factor an algebraic expression by taking out its greatest common factor.

Learn the idea

A common factor divides every term in an expression. The greatest common factor is the largest factor shared by all terms.

Use the distributive property in reverse: $ab + ac = a(b + c)$. Check by expanding your answer.

Worked example

$18x + 24 = 6(3x + 4)$, because 6 is the greatest common factor of 18 and 24.

Try it

1. Factorize $12x + 18$.

Answer: _____

2. Factorize $14a - 21$.

Answer: _____

3. Factorize $9m + 6$.

Answer: _____

4. Factorize $8x^2 + 12x$.

Answer: _____

Lesson 7 Direct Proportion

Identify directly proportional quantities and use a constant unit rate to solve problems.

Learn the idea

Two quantities are directly proportional when their ratio is constant. If y is directly proportional to x , then $y = kx$ for a constant k .

A table of directly proportional values has the same value of y/x in every row. Its graph is a straight line through the origin.

Worked example

Four notebooks cost 6 USD. One notebook costs $6 \div 4 = 1.5$ USD, so 10 notebooks cost $10 \times 1.5 = 15$ USD.

Try it

1. At 2.4 USD per kilogram, what is the cost of 3 kg?

Answer: _____

2. If $y = 5x$, find y when $x = 8$.

Answer: _____

3. Six bottles cost 9 USD. What do 10 bottles cost at the same rate?

Answer: _____

4. Are the pairs (2, 8), (4, 16), and (6, 24) directly proportional? Give the constant.

Answer: _____

Lesson 8 Calculating with Algebraic Expressions

Combine like terms, use the distributive property, and evaluate an expression for a given value.

Learn the idea

Like terms have the same variable part. Add or subtract their coefficients, and keep the variable part unchanged.

To evaluate an expression, substitute the given value for each variable, then follow the order of operations.

Worked example

$3(2x + 1) - x = 6x + 3 - x = 5x + 3$. When $x = 2$, the value is 13.

Try it

1. Simplify $7a - 2a + 4$.

Answer: _____

2. Expand $4(y - 3)$.

Answer: _____

3. Evaluate $2m + 5$ when $m = -3$.

Answer: _____

4. Simplify $2x + 3x - 4 + 7$.

Answer: _____

Lesson 9 Equations of the Form $ax = b$

Solve a one-step equation $ax = b$ by dividing both sides by the same non-zero number.

Learn the idea

An equation remains balanced when the same operation is applied to both sides. For $ax = b$, divide both sides by a , where a is not zero.

Check a solution by substituting it into the original equation.

Worked example

Solve $-6x = 42$. Divide both sides by -6 : $x = -7$. Check: $-6 \times (-7) = 42$.

Try it

1. Solve $8x = 56$.

Answer: _____

2. Solve $(1/3)x = 5$.

Answer: _____

3. Solve $-4x = -28$.

Answer: _____

4. Eight equal packets contain 56 pencils altogether. How many pencils are in each packet?

Answer: _____

Lesson 10 Coordinates and Geometric Loci

Read coordinates in the plane and describe simple loci using distance.

Learn the idea

A point is written (x, y) : move x units horizontally from the origin, then y units vertically. Positive x points right; positive y points up.

The points at a fixed distance from a fixed point form a circle. Points equidistant from two fixed points lie on the perpendicular bisector of the segment joining them. Points equidistant from the two sides of an angle lie on its angle bisector.

Worked example

$A(-2, 0)$ and $B(4, 0)$ lie on a horizontal line. Their midpoint has x -coordinate 1, so the perpendicular bisector is the vertical line $x = 1$.

Try it

1. Which quadrant contains $C(-3, 5)$?

Answer: _____

2. Give the coordinates of a point on the y -axis 4 units below the origin.

Answer: _____

3. What is the locus of points equidistant from A and B ?

Answer: _____

4. What is the locus of points 5 cm from O ?

Answer: _____

Lesson 11 Constructing Perpendicular and Angle Bisectors

Construct perpendicular bisectors and angle bisectors with a compass and ruler.

Learn the idea

To construct the perpendicular bisector of AB, draw equal-radius arcs from A and B using a radius greater than half of AB. Join the two arc intersections.

To bisect an angle, draw an arc from the vertex to meet both sides. From those two points draw equal-radius arcs that intersect inside the angle. Join the vertex to that intersection.

Worked example

If P lies on the perpendicular bisector of AB and $PA = 6$ cm, then $PB = 6$ cm as well.

Try it

1. What does every point on the perpendicular bisector of AB have in common?

Answer: _____

2. Name the two arc intersections used to draw the perpendicular bisector.

Answer: _____

3. If P is on an angle bisector and is 3 cm from one side, how far is it from the other side?

Answer: _____

4. After marking two equal-radius arcs from the sides of an angle, what do you do next?

Answer: _____

Lesson 12 Congruent Triangles and Angles on Parallel Lines

Recognize triangle congruence and use angle relationships created by a transversal.

Learn the idea

Congruent triangles have equal corresponding sides and angles. Common tests include SSS, SAS, and ASA; match the corresponding vertices in the same order.

When a transversal cuts parallel lines, corresponding angles and alternate interior angles are equal.

Co-interior angles on the same side of the transversal add to 180 degrees.

Worked example

If a co-interior angle is 118 degrees, the other co-interior angle is $180 - 118 = 62$ degrees. An alternate interior angle to 118 degrees is 118 degrees.

Try it

1. If a corresponding angle is 72 degrees, find the matching corresponding angle.

Answer: _____

2. Solve $3x + 10 = 70$ for x .

Answer: _____

3. One co-interior angle is 115 degrees. Find the other.

Answer: _____

4. Two triangles have matching side lengths 3 cm, 4 cm, and 5 cm. Which congruence test applies?

Answer: _____

Lesson 13 Representing Cubes and Rectangular Prisms

Describe and sketch plane representations and nets of a cube and a rectangular prism.

Learn the idea

A cube has 6 square faces, 12 edges, and 8 vertices. A rectangular prism also has 6 faces, 12 edges, and 8 vertices; opposite faces are equal rectangles.

A net is a flat arrangement of the solid's faces that can fold into the solid. A drawing of a solid on paper uses visible edges to suggest depth.

Worked example

A rectangular prism with side lengths 2 cm, 3 cm, and 5 cm has three pairs of equal faces: 2 by 3, 2 by 5, and 3 by 5.

Try it

1. How many faces, edges, and vertices does a cube have?

Answer: _____

2. How many faces does a rectangular prism have?

Answer: _____

3. What is a net?

Answer: _____

4. List the three face sizes of a 2 cm by 3 cm by 5 cm rectangular prism.

Answer: _____

Lesson 14 Translations

Move every point of a figure by the same vector while preserving lengths and angles.

Learn the idea

A translation moves each point the same distance in the same direction. It preserves size, shape, lengths, and angles.

On a coordinate plane, translation by (a, b) maps (x, y) to $(x + a, y + b)$.

Worked example

Translate $A(2, -1)$ by vector $(-3, 4)$: $A' = (2 - 3, -1 + 4) = (-1, 3)$.

Try it

1. Translate $B(-2, 5)$ by vector $(4, -3)$.

Answer: _____

2. Find the vector that maps $(1, 1)$ to $(5, 1)$.

Answer: _____

3. Does a translation preserve the length of a segment?

Answer: _____

4. Translate $C(0, -2)$ by vector $(-2, 2)$.

Answer: _____

Lesson 15 Relative Frequencies and Graphs

Calculate relative frequencies and represent discrete data with a bar graph or frequency polygon.

Learn the idea

The relative frequency of a category is its frequency divided by the total number of observations. Write it as a fraction, decimal, or percentage.

A bar graph compares category frequencies. A frequency polygon plots each value against its frequency and joins consecutive points with straight segments. Relative frequencies add to 1, or 100 percent.

Worked example

For counts 2, 5, 6, and 3, the total is 16. The relative frequency of the second count is $5/16 = 0.3125 = 31.25$ percent.

Try it

1. Six of 30 students chose blue. Find the relative frequency as a decimal and a percentage.

Answer: _____

2. Counts are 4, 10, and 6. Find each relative frequency.

Answer: _____

3. If a category has relative frequency 0.24 in a group of 25, how many observations are in it?

Answer: _____

4. A discrete data set has values 0, 1, 2, 3 with frequencies 2, 5, 6, 3. Give the four points for its frequency polygon.

Answer: _____

Cumulative Review A Number and Algebra

Try the questions without using the lesson pages. Show your working for questions that need more than one step.

1. Write 126 as a product of prime factors.

Answer: _____

2. Reduce $\frac{64}{96}$.

Answer: _____

3. Write $\frac{13}{40}$ as a decimal.

Answer: _____

4. Calculate $-7 - (-12)$.

Answer: _____

5. Calculate $(-9) \times (-4)$.

Answer: _____

6. Evaluate $2^4 + 3^2$.

Answer: _____

7. Factorize $18x - 24$.

Answer: _____

8. Five metres of fabric cost 17.5 USD. Find the cost of 8 metres at the same rate.

Answer: _____

9. Simplify $4a + 3 - 2a + 5$.

Answer: _____

10. Solve $7x = 63$.

Answer: _____

11. Solve $0.25x = 8$.

Answer: _____

12. Six identical items cost 9 USD. Find the cost of 10 items.

Answer: _____

Cumulative Review B Geometry and Data

Use a ruler and compass for construction questions. Show your reasoning when using an angle relationship.

1. Which quadrant contains $A(-2, 3)$?

Answer: _____

2. Describe the locus of points 4 cm from O.

Answer: _____

3. What is the locus of points equidistant from A and B?

Answer: _____

4. If a point is on an angle bisector and is 5 cm from one side, how far is it from the other side?

Answer: _____

5. An alternate interior angle is 115 degrees. Find its matching angle.

Answer: _____

6. One co-interior angle is 78 degrees. Find the other.

Answer: _____

7. Two triangles have matching sides 3, 4, and 5 cm. State a congruence test.

Answer: _____

8. How many faces, edges, and vertices does a cube have?

Answer: _____

9. How many faces, edges, and vertices does a rectangular prism have?

Answer: _____

10. Translate $(-1, 4)$ by vector $(3, -2)$.

Answer: _____

11. Nine of 30 students chose a category. Find its relative frequency as a percentage.

Answer: _____

12. Counts are 2, 4, and 6. Find the total and the three relative frequencies.

Answer: _____

Short Test

Suggested time: 25 minutes. Work independently, then check the answer key.

1. Is 29 prime or composite?

Answer: _____

2. Write 90 as a product of prime factors.

Answer: _____

3. Reduce $\frac{36}{48}$.

Answer: _____

4. Write $\frac{11}{25}$ as a decimal.

Answer: _____

5. Calculate $-4 - (-9)$.

Answer: _____

6. Calculate $(-6) \times 7$.

Answer: _____

7. Evaluate 5^3 .

Answer: _____

8. Factorize $20x + 30$.

Answer: _____

9. Four metres cost 10 USD. What do 7 metres cost at the same rate?

Answer: _____

10. Simplify $2x + 7x - 3$.

Answer: _____

11. Solve $-5x = 35$.

Answer: _____

12. Which axis contains the point $(0, -3)$?

Answer: _____

13. One co-interior angle is 118 degrees. Find the other.

Answer: _____

14. Translate $P(-2, 1)$ by vector $(5, -3)$.

Answer: _____

15. Three of 20 observations have a given category. Find the relative frequency as a percentage.

Answer: _____

Answer Key

Check each answer after completing the practice. For a mistake, return to the lesson and solve a similar question again.

Lesson 1 Prime Numbers and Prime Factorization

1. 1: neither; 2: prime; 15: composite; 23: prime.
2. $72 = 2^3 \times 3^2$.
3. $105 = 3 \times 5 \times 7$.
4. No. $91 = 7 \times 13$.

Lesson 2 Reducing Fractions

1. $\frac{3}{5}$.
2. $\frac{4}{5}$.
3. $\frac{7}{9}$.
4. $\frac{3}{8}$.

Lesson 3 Writing Fractions as Decimals

1. 0.375.
2. 0.36.
3. 1.75.
4. 0.55.

Lesson 4 Subtracting and Multiplying Integers

1. 20.
2. -13.
3. -42.
4. 36.

Lesson 5 Powers with Positive Integer Exponents

1. 64.
2. 49.
3. $\frac{1}{8}$.
4. 17.

Lesson 6 Common Factors and Factorization

1. $6(2x + 3)$.
2. $7(2a - 3)$.
3. $3(3m + 2)$.
4. $4x(2x + 3)$.

Lesson 7 Direct Proportion

1. 7.2 USD.
2. 40.
3. 15 USD.
4. Yes. $y/x = 4$ for each pair, so $k = 4$.

Lesson 8 Calculating with Algebraic Expressions

1. $5a + 4$.
2. $4y - 12$.
3. -1.
4. $5x + 3$.

Lesson 9 Equations of the Form $ax = b$

1. $x = 7$.
2. $x = 15$.
3. $x = 7$.
4. 7 pencils.

Lesson 10 Coordinates and Geometric Loci

1. Quadrant II.
2. (0, -4).
3. The perpendicular bisector of segment AB.
4. A circle with centre O and radius 5 cm.

Lesson 11 Constructing Perpendicular and Angle Bisectors

1. It is the same distance from A and B.
2. The two intersections of the equal-radius arcs centred at A and B.
3. 3 cm.
4. Draw equal-radius arcs from the two points on the sides; join their intersection to the angle's vertex.

Lesson 12 Congruent Triangles and Angles on Parallel Lines

1. 72 degrees.
2. $x = 20$.
3. 65 degrees.
4. SSS.

Lesson 13 Representing Cubes and Rectangular Prisms

1. 6 faces, 12 edges, 8 vertices.
2. 6 faces.
3. A flat arrangement of faces that folds into a solid.
4. 2 by 3, 2 by 5, and 3 by 5 cm.

Lesson 14 Translations

1. $B' = (2, 2)$.
2. $(4, 0)$.
3. Yes.
4. $C' = (-2, 0)$.

Lesson 15 Relative Frequencies and Graphs

1. 0.2; 20 percent.
2. 0.2, 0.5, 0.3.
3. 6 observations.
4. $(0, 2)$, $(1, 5)$, $(2, 6)$, $(3, 3)$.

Cumulative Reviews and Short Test

Review A

1. $126 = 2 \times 3^2 \times 7$.
2. $\frac{2}{3}$.
3. 0.325.
4. 5.
5. 36.
6. 25.
7. $6(3x - 4)$.
8. 28 USD.
9. $2a + 8$.
10. $x = 9$.
11. $x = 32$.
12. 15 USD.

Review B

1. Quadrant II.
2. A circle centred at O with radius 4 cm.
3. The perpendicular bisector of AB.
4. 5 cm.
5. 115 degrees.
6. 102 degrees.
7. SSS.
8. 6 faces, 12 edges, 8 vertices.
9. 6 faces, 12 edges, 8 vertices.
10. (2, 2).
11. 30 percent.
12. Total 12; relative frequencies $\frac{1}{6}$, $\frac{1}{3}$, and $\frac{1}{2}$.

Short Test

1. Prime.
2. $90 = 2 \times 3^2 \times 5$.
3. $\frac{3}{4}$.
4. 0.44.
5. 5.
6. -42.
7. 125.
8. $10(2x + 3)$.
9. 17.5 USD.
10. $9x - 3$.
11. $x = -7$.
12. The y-axis.
13. 62 degrees.
14. $P' = (3, -2)$.
15. 15 percent.

Curriculum Reference

The topic selection follows the Seventh Year scope and sequence in the English Mathematics curriculum for the Intermediate Level published by the Centre for Educational Research and Development (CRDP), Lebanon.

The covered Grade 7 areas include prime numbers and factorization; reducing fractions and decimal writing; integer subtraction and multiplication; positive integer powers; common-factor factorization; direct proportion; algebraic expressions; equations $ax = b$; coordinates and geometric loci; constructions; congruent triangles and angles on parallel lines; representations of cubes and rectangular prisms; translations; relative frequencies; bar graphs; and frequency polygons.

Official source: [CRDP Lebanon English Mathematics curriculum](#)

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